



Deep Voice - Applied ML Engineer: Technical Assignment

Objective

The goal of this assignment is to evaluate your practical understanding of machine learning pipelines, model training practices, and result analysis. You will build and compare two different classification approaches to detect North Atlantic Right Whale upcalls in noisy ocean acoustic data.

The Dataset

We are using a legacy dataset from a Kaggle competition as our toy problem.

- **Data Link:** [The ICML 2013 Whale Challenge - Right Whale Redux](#)
- *Note: You will need to process the raw audio files appropriately depending on the model architectures you choose.*

The Task

Please implement an end-to-end training and evaluation pipeline that does the following:

1. **Baseline Image Classifier:** Convert the audio data into spectrograms and fine-tune an off-the-shelf image-based classifier (e.g., ResNet-18, EfficientNet, or similar) to classify the presence of a whale call.
2. **Audio Foundation Model:** Adapt and evaluate a specialized audio foundation model for the same classification task. You can choose to build your classifier based on either:
 - [Perch \(ONNX\)](#)
 - [BEATs \(Microsoft Unilm\)](#)
3. **Comparison & Analysis:** Compare the performance, training dynamics, and practical trade-offs between the two approaches.

Deliverables (Due April 19th) Please submit the following:

1. **The Code:** A link to a GitHub repository (or a zipped folder) containing your code. It should be clean, readable, and structured like a standard ML pipeline.
2. **The Report:** A summary report documenting your preprocessing choices, training mechanisms, evaluation metrics, and a comparison of the results.
 - *Format:* We highly prefer a **Weights & Biases (W&B) Report** link, but a well-formatted PDF is also perfectly acceptable.



What We Are Looking For

During our 45-minute follow-up call, we will use your submission to anchor our discussion. We are not just looking for the highest accuracy score. We will evaluate:

- **Engineering Best Practices:** How you structure your data loading, training loops, and validation.
- **Problem Solving:** How you handle the transition from raw audio to model-ready inputs for different architectures.
- **Analysis:** Your ability to read training curves, interpret metrics (e.g., handling false positives vs. false negatives in a conservation context), and understand *why* one model might outperform the other.
- **AI Tooling:** You are fully allowed and encouraged to use AI coding assistants to write this pipeline. We want to see how you orchestrate the tools to get high-quality results efficiently.